Disease Simulation Report

Deggendorf Institute of Technology
High Performance Computing and Quantum Computing
Module: HPC Programming Lab

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1 Introduction

Infectious diseases can spread rapidly, often straining healthcare and government systems. Traditional models, based on fixed assumptions like constant transmission rates, struggle to capture the complex, dynamic nature of real-world disease spread.

This project uses a simulation-based SIR model to explore how diseases propagate through a population. By adjusting parameters such as vaccination rate, transmission rate, and disease duration, the simulation reveals trends in infection peaks, susceptibility changes, and the effects of vaccination and recovery over time.

2 Methods

2.1 Model Description

The simulation uses the classical Susceptible-Infected-Recovered (SIR) model. The population is divided into three compartments:

- **Susceptible (S)**: Individuals who can contract the disease.
- **Infected (I)**: Individuals who can transmit the disease.
- **Recovered or Vaccinated (R)**: Individuals who have gained immunity to infection.

2.2 Simulation Logic

The simulation starts by vaccinating a specified percentage of the population. From the remaining population, one individual is marked as infected and the rest as susceptible. Each infected person randomly contacts six individuals per day. If a contacted person is susceptible, they may become infected based on a specified transmission rate. After a defined infectious period, infected individuals recover.

2.3 Parameter Configuration

Key parameters are adjustable via an INI configuration file:

- Population size: 30000
- Initial infected: 1
- Disease duration: 4 days
- Transmission rate (β): 0.25

2.4 Implementation Tools

- **Language**: C++
- **Output**: Daily and statistical S, I, R values stored in CSV files
- **Visualization**: Matplotlib
- **Testing**: Doctest framework
- **Build**: CMake, Docker

3 Results

The simulation demonstrates how an outbreak evolves under specific parameter settings. The data presented is based on the **average of five simulation runs** to ensure consistency and reliability.

3.1 Graphical Trends

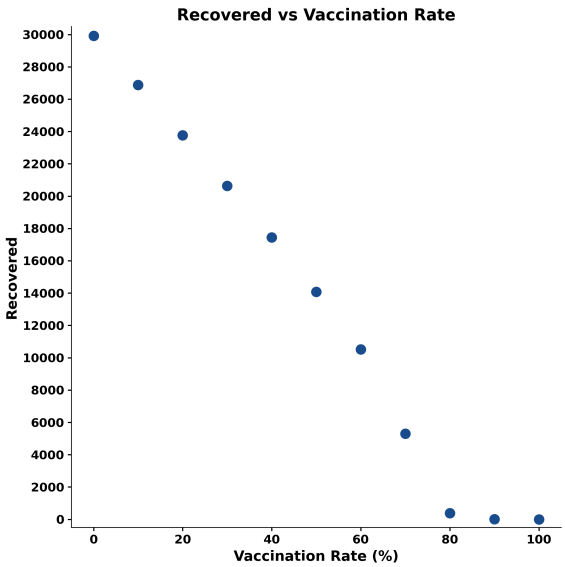

Figure 1: Simulation for D = 4, C = 6, β = 0.25

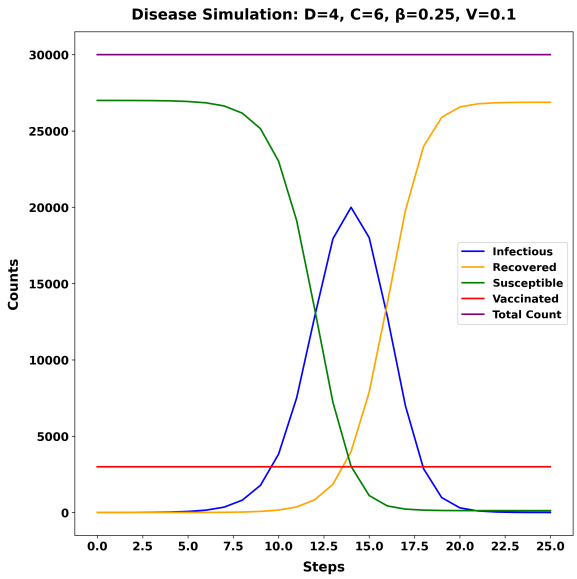

Figure 2: Timestep plot for D = 4, C = 6, β = 0.25 with vaccination rate $V = 0.1$

3.2 Herd Immunity

In the SIR simulation, the basic reproduction number R_0 is calculated as:

$$R_0 = D \cdot \beta \cdot C$$

For this model, where $C = 6$, $\beta = 0.25$, and $D = 4$, we get:

$$R_0 = 4 \cdot 0.25 \cdot 6 = 6.0$$

Based on the Herd Immunity Graph[1], the herd immunity threshold for $R_0 = 6$ is approximately 85%.

3.3 Epidemic Behavior

- The number of infected individuals decreases with higher vaccination rates. (Figure 1)
- Once the vaccination rate reaches around 85%, no further infections are observed—indicating successful herd immunity. (Figure 1)
- Infections rise sharply, peaking around day 14 of the simulation. (Figure 2)
- After the peak, recoveries begin to outnumber new infections. (Figure 2)
- The outbreak eventually subsides as the susceptible population shrinks. (Figure 2)

4 Discussion

The simulation shows that outbreaks can grow quickly when the disease spreads easily or lasts longer. Starting with some people already vaccinated helps slow the spread by reducing the number of people who can get infected.

4.1 Limitations

- Everyone is assumed to have an equal chance of contact.
- No interventions like lockdowns or isolation are included.
- Differences in age, location, or other factors are not considered.

4.2 Future Work

Future improvements could include:

- Include virus mutation and its effects on transmission or recovery.
- Simulate contact patterns using real-life social network structures.
- Incorporate dynamic interventions

5 Conclusion

This project used a simple simulation to show how diseases spread and how vaccination helps. The findings show that:

- Vaccination lowers infection peaks and slows the outbreak.
- Infections grow quickly at first, then drop as people recover.
- A high vaccination rate (around 85%) can prevent further spread and lead to herd immunity.

The model highlights the importance of early vaccination and helps in understanding how outbreaks can be managed.

[1] https://commons.wikimedia.org/wiki/File:Herd_immunity_threshold_vs_r0.svg